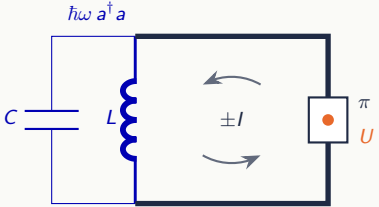
Zero-point flux fluctuations switch a quantum-dot Josephson loop from 0 to π

An exact electron–photon solution of a π loop sharing its inductance with a cavity mode

---

Group seminar

The story in one slide



A BCS ring closed by an interacting quantum dot, threaded by the quantized flux of one LC mode. Solved *exactly* (DMRG on electrons $\otimes$ photon).

1. **The vacuum of the mode controls the junction.** Its zero-point flux moves the singlet–doublet (0–π) boundary: at $Z \simeq 2\,\mathrm{k}\Omega$ ($g \simeq 0.5$) a 0 junction becomes a π junction. Adiabatic mechanism (curvatures of $E_s(\varphi)$) $\rightarrow$ Franck–Condon mechanism ($t_J \rightarrow t_J e^{-g^2/2}$).
2. **Full-DMRG phase diagram in (U, g).** For $U > U_c$, $g > g_c(U)$ the mode condenses ($\langle a \rangle \neq 0$ = a persistent current). This is the classical screening instability $\beta > 1$ of a π loop; the exact ground state is a cat of the two current directions, and zero-point motion pushes the threshold above $\beta = 1$.
3. **The product-state mean field orders where the exact solution does not**: for $E_J \lesssim 3\hbar\omega$ and at $U \simeq U_c$, where a static moment replaces the Kramers doublet.

Outline

1. Background and what is new
2. The microscopic model
3. Theory: the two limits
4. Methods
5. Results
6. Where to look
7. Summary

Background and what is new

π loops, photon condensation, and the no-go theorem

Known since 1977: a loop closed by a π junction carries a spontaneous current when $L > \Phi_0/2\pi I_c$ [Bulaevskii–Kuzii–Sobyanin 1977; Tsuei–Kirtley 2000; Bauer 2004]; with a small C the two directions form a flux qubit without external field [Ioffe 1999; Kawabata 2006; Feofanov 2010]. **Cavity QED version:** a mode coupling through its *flux* evades the no-go theorem for superradiance and condenses a static field $\langle a \rangle \neq 0$ [Nataf–Ciuti 2010; Bamba 2016; Andolina 2020; Mercurio 2024] – the condition is the magnetostatic instability of the matter.

What was missing

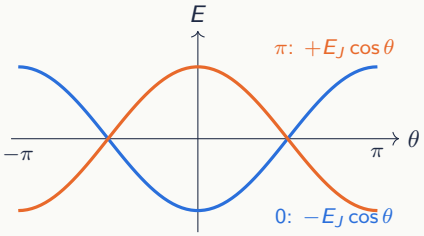
In all of these the junction is a *fixed* function $E(\varphi)$, or electrons and mode are decoupled in a product state [Varrica 2026].

This work

A quantum-dot junction is not a fixed $E(\varphi)$: it is π only in its doublet state, and whether it is 0 or π is decided by U, Γ, Δ on a scale of a few GHz – comparable to $\hbar\omega$.

$\Rightarrow$ treat electrons and quantized flux on the same footing and solve *exactly* by DMRG [cf. Passetti 2023; Bacciconi 2023].

Warm-up: 0 and π Josephson junctions



θ = gauge-invariant Cooper-pair phase difference.

- **0 junction:** minimum at $\theta = 0$, $E'' > 0$: an *inductance* $L_J = (\Phi_0/2\pi)^2/E_J$, diamagnetic.
- **π junction:** minimum at $\theta = \pi$; at $\theta = 0$ a *maximum*: a negative inductance, paramagnetic.
- Realizations: SFS, d-wave corners, and quantum dots with a local moment (this talk).

Classical π rf-SQUID

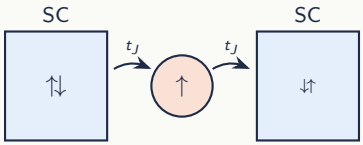
$U(\theta) = \frac{E_L}{2}\theta^2 + E_J \cos\theta, \quad E_L = \frac{(\Phi_0/2\pi)^2}{L}.$
$U''(0) = E_L - E_J < 0$ iff $\beta \equiv \frac{E_J}{E_L} = \frac{2\pi L I_c}{\Phi_0} > 1.$

Why a quantum dot with a local moment is a π junction

Anderson impurity between BCS leads. Two competing ground states of the ring:

- **singlet** (even fermion parity): the dot electron is screened/paired; a 0 junction, $E_2 < 0$;
- **doublet** (odd parity, Kramers pair): a Cooper pair crosses by fourth-order cotunnelling, one electron at a time, and arrives with its spins in reversed order: $c_\downarrow^\dagger c_\uparrow^\dagger = -c_\uparrow^\dagger c_\downarrow^\dagger \Rightarrow E_J \to -E_J$, a π junction, $E_2 > 0$.

[Glazman–Matveev 1989; Spivak–Kivelson 1991; Vecino 2003; Karrasch 2008]; seen in nanowire and CNT dots [van Dam 2006; Cleuziou 2006; Delagrange 2015]; read out via a transmon/resonator [Bargerbos 2022; Hays 2021].



4 hops, intermediate energies $\sim U, \Delta$

The 0–π transition is a *parity* transition, singlet $\leftrightarrow$ doublet, tuned by U/Γ, Γ/Δ and the gate. Fermion parity is conserved $\Rightarrow$ the two sectors can be solved separately and compared.

The microscopic model

Step 1: flux enters as a Peierls phase

Tight binding in a vector potential:

$$c_{j+1}^{\dagger} c_j \longrightarrow e^{i\theta_j}\, c_{j+1}^{\dagger} c_j, \qquad \theta_j = \frac{e}{\hbar} \int_j^{j+1} \mathbf{A} \cdot d\mathbf{l}.$$

For a *static* phase, $c_j \to e^{i\chi_j} c_j$ moves phase between bonds (with $\Delta_j \to \Delta e^{-2i\chi_j}$ [Dmytruk–Schirò 2021]); only the loop sum is physical:

$$\varphi \equiv \sum_j \theta_j = \frac{e}{\hbar} \oint \mathbf{A} \cdot d\mathbf{l} = \frac{e}{\hbar}\, \Phi.$$

Dictionary (keep this in mind!)

electron phase	$\varphi = e\Phi/\hbar$
Cooper-pair phase	$\theta = 2\varphi$
$\Phi = \Phi_0 = h/2e$	$\varphi = \pi$
$\Phi = \Phi_0/2$	$\varphi = \pi/2$
current	$I = -\partial E/\partial\Phi = -\frac{e}{\hbar}\,\partial_\varphi E$

For the quantized flux the placement is physics, not gauge

Moving a fraction λ of gX to other bonds ($W = e^{-igX\lambda N}$) gives $a^\dagger a \to a^\dagger a + g\lambda NP + g^2\lambda^2 N^2$: a *capacitive* coupling of the displaced charge N to the charge P of the mode [Dmytruk–Trif–Simon 2016]. One-bond vertex = the voltage of the mode drops across the junction (the only high-impedance element). Neglected: $O(\hbar\omega g^2/E_A)$.

Step 2: quantize the flux of the LC mode

$$H_{LC} = \frac{Q^2}{2C} + \frac{\Phi^2}{2L} = \hbar\omega\left(a^\dagger a + \tfrac{1}{2}\right)$$

$$\Phi = \Phi_{\mathrm{zpf}}\,(a + a^\dagger), \qquad \Phi_{\mathrm{zpf}} = \sqrt{\hbar Z/2}$$

($Z = \sqrt{L/C}$). The loop phase becomes an operator:

$$\varphi \;\to\; \varphi_{\mathrm{cl}} + g\,(a + a^\dagger), \qquad g = \frac{e\,\Phi_{\mathrm{zpf}}}{\hbar}$$

g in circuit language

$g^2 = \dfrac{e^2 Z}{2\hbar} = \dfrac{\pi}{4}\dfrac{Z}{R_Q}$, $R_Q = h/4e^2 \simeq 6.45\,\mathrm{k\Omega}$.
$Z = 50\,\Omega \Rightarrow g = 0.08$; $Z = 1\,\mathrm{k\Omega} \Rightarrow g = 0.35$;
$Z = 2\,\mathrm{k\Omega} \Rightarrow g = 0.5$.

The junction hopping becomes a photon displacement:

$$-t_J\, e^{i\varphi_{\mathrm{cl}}}\, \underbrace{e^{ig(a+a^\dagger)}}_{D}\, c^\dagger_{1\sigma} c_{0\sigma} + \mathrm{h.c.}$$

- each tunnelling electron kicks the oscillator;
- expanding: $e^{igX} = 1 + igX \underbrace{- \frac{g^2}{2} X^2}_{A^2\text{ term}} + \ldots$
  $\Rightarrow$ diamagnetic terms included to all orders [Li 2020; Dmytruk–Schirò 2021];
- vacuum average: $\langle 0|D|0\rangle = e^{-g^2/2}$ (Franck–Condon factor) – remember this for the anti-adiabatic limit.

The full Hamiltonian

$$H = \hbar\omega\, a^\dagger a \underbrace{- t \sum_{r=1}^{L-2} \sum_{\sigma} \left(c^\dagger_{r+1\sigma} c_{r\sigma} + \text{h.c.} \right) + \Delta \sum_{r=1}^{L-1} \left(c^\dagger_{r\uparrow} c^\dagger_{r\downarrow} + \text{h.c.} \right)}_{s\text{-wave BCS chain (the ``leads'')}}$$

$$+ \underbrace{\epsilon_d\, n_0 + U\, n_{0\uparrow} n_{0\downarrow}}_{\text{Anderson dot, site 0}} - t_J \sum_{\sigma} \left(c^\dagger_{0\sigma} c_{L-1\,\sigma} + \text{h.c.} \right)$$

$$- t_J \sum_{\sigma} \left(e^{i\varphi_{\rm cl}}\, e^{ig(a+a^\dagger)}\, c^\dagger_{1\sigma} c_{0\sigma} + \text{h.c.} \right) \equiv \hbar\omega\, a^\dagger a + H_{\rm ring}\left[\varphi_{\rm cl} + g(a + a^\dagger)\right]$$

Parameters (energies in Δ, currents in $e\Delta/\hbar$)

$\Delta = t$, $t_J = 0.5\Delta$ ($\Gamma = 2t_J^2/t = \Delta/2$), $\epsilon_d = -U/2$ (particle–hole symmetric), $L = 10 > \xi \simeq t/\Delta$ (finite-size and h/e corrections $\leq 2\%$), $\varphi_{\rm cl} = 0$.
Phase diagram: $\hbar\omega = E_J(U = 2\Delta)/10 = 7.39 \times 10^{-4}\Delta$.
Flux-driven switch: $\hbar\omega = 0.01$–$3\,\Delta$.

$\varphi_{\rm cl}$: a static flux, used only as the $\Phi_0/2$ bias of a weak link (backup) or as an infinitesimal seed (10^{-3}) to select a current direction. Conserved: fermion parity s and S^z.

One function per parity sector: $E_s(\varphi)$, and an exact sum rule

Ground-state energy of $H_{\rm ring}$ at fixed phase, in each sector $s = \text{even/odd}$:

$$E_s(\varphi) = \sum_{m \geq 0} E_m^{(s)} \cos m\varphi$$

- Only cosines: $H[\varphi_{\rm cl} = 0]$ is real $\Rightarrow E_s(\varphi) = E_s(-\varphi)$ *exactly*.
- $m = 2, 4, \ldots$: Cooper pairs ($h/2e$); $m = 1$: one electron around the ring, $\propto e^{-L/\xi}$ [Loss–Goldbart 1991].
- singlet: $E_2 < 0$ (0 junction, $E''(0) > 0$); doublet: $E_2 > 0$ (π junction, $E''(0) < 0$).
- $E_J \equiv E_2$, $I_c = 2eE_J/\hbar$ (pure π: $I = I_c \sin 2\varphi$).

Current = displacement of the mode

$I = -\frac{e}{\hbar}\partial H/\partial\varphi_{\rm cl}$ (particle current through the dot bond, dressed by D). From $\langle [H, P]\rangle = 0$, $P = i(a^\dagger - a)$:

$$[a^\dagger a, P] = iX, \qquad [H_{\rm ring}, P] = 2ig\,\partial_\varphi H_{\rm ring}$$

$$\Rightarrow \quad \langle I\rangle = \frac{e}{\hbar}\,\frac{\hbar\omega}{2g}\,\langle a + a^\dagger\rangle = \frac{\langle \Phi\rangle}{L}$$

in *every* eigenstate (use $2\Phi_{\rm zpf}^2 = \hbar\omega L$).

Both sides are computed independently (fermionic vs photonic observable): this sum rule is the acceptance test of every DMRG state.

Theory: the two limits

Limit 1 ($\hbar\omega \ll E_A$): Born–Oppenheimer $\to$ the circuit model

Electrons follow the flux adiabatically if $\hbar\omega \ll E_A$ (excitation gap of the ring at fixed phase and parity). Flux quadratures $X = a + a^\dagger$, $P = i(a^\dagger - a)$, $[X, P] = 2i$, $\hbar\omega a^\dagger a = \frac{\hbar\omega}{4}(X^2 + P^2) - \frac{\hbar\omega}{2}$. Ansatz $|\Psi\rangle = \int dX\, \chi(X)\, |X\rangle \otimes |\psi_0^{(s)}(\varphi_{\rm cl} + gX)\rangle$:

$$H_{\rm BO}^{(s)} = \frac{\hbar\omega}{4}\left(P^2 + X^2\right) + E_s(\varphi_{\rm cl} + gX)$$

A 1D Schrödinger problem for the flux, *per parity sector*; photon treated exactly (all Fock states); full $E_s(\varphi)$ from DMRG at $g = 0$ (singlet: spline through 25 phases, it has a cusp). Called the circuit model below.

Cooper-pair phase $\theta = 2gX$, charge $n = P/4g$, $[\theta, n] = i$

$\frac{\hbar\omega}{4} X^2 = \frac{1}{2} E_L \theta^2$, $E_L = \dfrac{\hbar\omega}{8g^2}$;
$\frac{\hbar\omega}{4} P^2 = 4E_C n^2$, $E_C = \hbar\omega g^2$
(check: $\sqrt{8E_C E_L} = \hbar\omega$)

$$H_{\rm BO} = 4E_C\, n^2 + \tfrac{1}{2} E_L\, \theta^2 + E_s(\theta/2)$$

= an rf-SQUID [Friedman 2000; Kawabata 2006], a fluxonium [Manucharyan 2009] whose Josephson element is the dot. Two knobs: $\beta = E_J/E_L$ and $E_J/\hbar\omega$. Small g: heavy, classical flux ($E_L/E_C = 1/8g^4$).

Landau theory of the condensate: $\beta > 1$

Classical energy (drop P):

$$\mathcal{F}(X) = \frac{\hbar\omega}{4}X^2 + E(gX),$$

$$\mathcal{F}''(0) = \frac{\hbar\omega}{2} + g^2E''(0) = \frac{\hbar\omega}{2}(1-\beta)$$

Instability criterion

$$\beta \equiv \frac{2g^2|E''(0)|}{\hbar\omega} = \frac{8g^2E_J}{\hbar\omega} = \frac{2\pi LI_c}{\Phi_0} > 1$$

with $E''(0) < 0$: needs the paramagnetic response of a π loop ($\beta = E_J/E_L$). The textbook screening parameter [BKS 1977]; the superradiance condition of [Bamba 2016]; the single-loop magnetostatic criterion of [Andolina 2020].

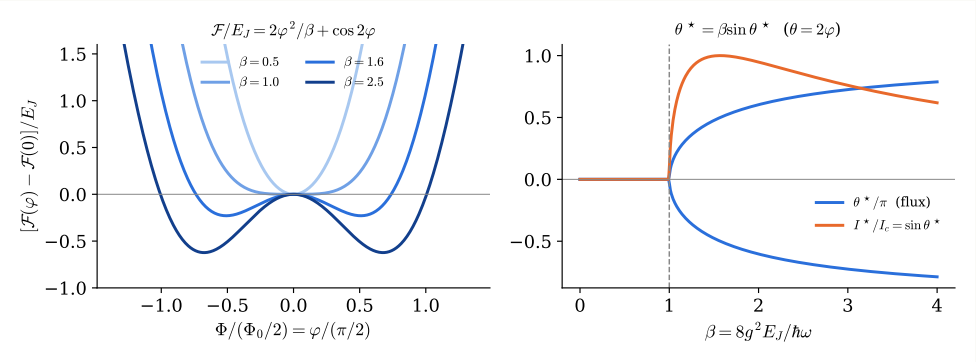
Numbers, doublet at $U = 2\Delta$

$E_J = E_2 = 7.39 \times 10^{-3}\Delta$, $\hbar\omega = E_J/10$
$\beta = 1 \;\Rightarrow\; g_c^{\mathrm{cl}} = \sqrt{\hbar\omega/8E_J} = 0.112$
full DMRG: $g_c = 0.13$ – above the classical line (why: next slides)

And in the singlet sector?

$E''_{\mathrm{even}}(0) = +0.20\Delta > 0$: the mode is *stiffened*,
$\omega' = \omega\sqrt{1 + 2g^2E''(0)/\hbar\omega}$ (up to $9.4\,\omega$ at $g = 0.4$);
ground state = squeezed vacuum,
$\langle n\rangle = \frac{1}{4}(\omega'/\omega + \omega/\omega') - \frac{1}{2} = 1.89$ vs DMRG 1.91. This is the inductive shift that reads out dot parity [Bargerbos 2022].

The classical double well and the pitchfork



$\mathcal{F}'(X) = 0$ with $\theta = 2gX$:

$$\frac{\hbar\omega}{2g}\,\varphi = 2gE_J \sin 2\varphi \;\Rightarrow\; \boxed{\theta^\star = \beta \sin\theta^\star}$$

$I^\star = I_c \sin\theta^\star$. Large β: $\theta^\star \to \pi$ (flux $\to \Phi_0/2$) and $I^\star \simeq \pi I_c/\beta \propto \hbar\omega/g^2$ *decreases*.

Below threshold: the soft mode

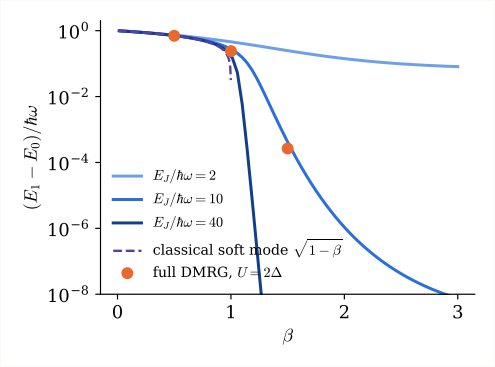
Expand to second order around $X = 0$ ($\beta < 1$):

$$H_{\mathrm{BO}} \simeq \frac{\hbar\omega}{4}\left(P^2 + (1-\beta)X^2\right)$$

- Excitation gap: $\boxed{\hbar\omega\sqrt{1-\beta}}$ (mode softening).
- Flux fluctuations grow: $\langle X^2\rangle = (1-\beta)^{-1/2}$.
- Photons in the ground state: $\langle n\rangle = \frac{1}{4}\left[(1-\beta)^{-\frac{1}{2}} + (1-\beta)^{\frac{1}{2}}\right] - \frac{1}{2}$.

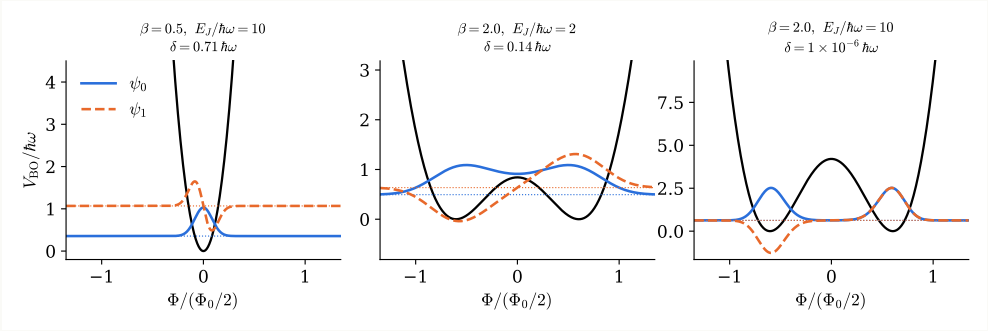
Check against full DMRG ($U = 2\Delta$)

$\beta = 0.5$: gap $\sqrt{0.5} = 0.707$ vs DMRG 0.70. $\beta = 1$: DMRG 0.24 (quartic regime). $\beta = 1.5$: DMRG resolves $\delta = 2.65 \times 10^{-4}\hbar\omega$ directly.



At $\beta \to 1$ the quadratic theory fails; above, the gap becomes the tunnel splitting and collapses exponentially.

Quantum fluctuations oppose the transition



Exact eigenstates of $H_{\rm BO}$ (pure π junction). For $\beta > 1$: $\psi_{0,1} \simeq (|\circlearrowleft\rangle \pm |\circlearrowright\rangle)/\sqrt{2}$ — cat states of macroscopic current, split by δ.

Zero-point fluctuations *restore* the symmetry [Bamba–Imoto 2017]: if the barrier is below $\sim \hbar\omega$ (middle) there is no current-carrying state at all.

How small is δ? An instanton estimate

Particle of mass $M = \hbar^2/8E_C$ in $U(\theta) = E_J\, u(\theta)$, $u = \frac{\theta^2}{2\beta} + \cos\theta$. WKB:

$$\delta \sim \hbar\omega\, e^{-S}, \quad S = \int_{-\theta^\star}^{\theta^\star} d\theta \sqrt{\frac{U - U_{\min}}{4E_C}}$$

Scaling

$$\frac{E_J}{4E_C} = \frac{E_J}{4g^2\hbar\omega} = \frac{2}{\beta}\left(\frac{E_J}{\hbar\omega}\right)^2 \Rightarrow \boxed{S = \frac{E_J}{\hbar\omega}\, s(\beta)}$$

$s(\beta) = \sqrt{2/\beta}\int d\theta\, \sqrt{u - u_{\min}}$.

Near threshold ($\epsilon = 1 - 1/\beta$): $u \simeq 1 - \frac{\epsilon}{2}\theta^2 + \frac{\theta^4}{24}$, minima at $\theta^2 = 6\epsilon$, depth $\frac{3}{2}\epsilon^2 \Rightarrow s \propto (\beta - 1)^{3/2}$.

Sharp transition only in a scaling limit

$E_J/\hbar\omega \to \infty$ at fixed β:

$$\frac{E_C}{E_J} = \frac{\beta}{8}\left(\frac{\hbar\omega}{E_J}\right)^2 \to 0,$$

flux becomes classical, $\delta \to 0$, pitchfork with mean-field exponents.

Cf. quantum Rabi model: QPT only for $\Omega/\omega \to \infty$ [Hwang–Puebla–Plenio 2015]. Finite $E_J/\hbar\omega$: a crossover; δ = macroscopic quantum tunnelling of the current.

Limit 2 ($\hbar\omega \gg E_A$): Lang–Firsov, the Franck–Condon reduction

When the mode is fast, it stays in its vacuum and the electrons see the *vacuum average* of the Peierls factor. Lang–Firsov transformation [Lang–Firsov 1963]:

$$D = e^{ig(a+a^\dagger)} = e^{-g^2/2}\, e^{iga^\dagger}\, e^{iga} \quad \Rightarrow \quad \langle 0|D|0\rangle = e^{-g^2/2}$$

$H \to H_{\rm ring}$ with $t_J \to t_J\, e^{-g^2/2}$ on the dressed bond, $+O(t_J^2/\hbar\omega)$

The zero-point flux reduces the level width of the dot by the Franck–Condon factor e^{-g^2} – on *one* of its two bonds.

Why this matters for 0 vs π

The singlet is bound by Γ (YSR/Kondo screening); the doublet is not. Reducing Γ favours the doublet: the boundary U_c moves *down*.

Adiabatic mechanism, for comparison

Zero-point motion samples $E_s(gX)$ around $\varphi = 0$: $\delta E_s \simeq \frac{1}{2} g^2 E_s''(0)\langle X^2\rangle$. Singlet curves *up* ($E''_{\rm even}(0) = +0.20\Delta$) and loses; doublet curves down and gains, then relaxes its flux towards $\Phi_0/2$. Also moves U_c *down*.

Neither limit is assumed in the DMRG; both are compared with it.

Methods

Three levels of description

	ansatz	photon	cost
Mean field (MF)	$\lvert\psi_{\mathrm{el}}\rangle \otimes \lvert\chi_{\mathrm{ph}}\rangle$, self-consistent; HF–BdG electrons [Varrica 2026; Andolina 2020]	product state	trivial
Circuit model	$H_{\mathrm{BO}}^{(s)}$ with the DMRG $E_s(\varphi)$ at $g = 0$ (fermions integrated out)	exact	1D Schrödinger
Lang–Firsov	H_{ring} with $t_J e^{-g^2/2}$	vacuum	BdG/DMRG of the ring
Full DMRG	MPS on electrons $\otimes$ photon, no approximation	exact	427 points $\approx$ 20 CPU h

Full DMRG is the result *and* the referee: the circuit model and Lang–Firsov are its two limits, mean field is the ansatz used in the literature for this circuit.

Mean field and its hidden assumptions

Product state $\Rightarrow$ two coupled problems:

el.: H_{ring} (HF–BdG) with $t_J \to t_J\,\alpha$, $\quad \alpha = \langle\chi|D|\chi\rangle$

ph.: $\hbar\omega a^\dagger a - \big(\eta\, D + \mathrm{h.c.}\big)$, $\quad \eta = t_J e^{i\varphi_{\mathrm{cl}}} \sum_\sigma \langle c^\dagger_{1\sigma} c_{0\sigma}\rangle$

$E_{\mathrm{MF}} = E_{\mathrm{el}} + E_{\mathrm{ph}} + 2\,\mathrm{Re}(\eta\alpha)$ (double counting)

For a coherent state $|X_0\rangle$: $\alpha = e^{igX_0}\, e^{-g^2/2}$.

(i) Not gauge covariant

The junction bond is shrunk by $e^{-g^2/2}$ and loses phase-*independent* kinetic energy, which the exact state recovers by following the flux.

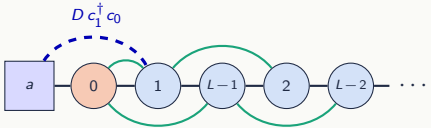
(ii) Self-trapping

A product state must pick one well: it breaks the $\mathbb{Z}_2$ by construction, even when the barrier is below the zero-point energy.

(iii) Static moment

Hartree–Fock replaces the Kramers doublet by a polarized moment: $U_c^{\mathrm{MF}} = 0.82\Delta$ for every g.

Full DMRG for electrons + photon, and why the protocol matters



ring in folded order: bonds of range ≤ 2

- TeNPy, two-site DMRG; photon = bosonic site, Fock cutoff 20–62; parity and S^z conserved; $\chi = 200$–400 (truncation error $< 2 \times 10^{-8}$).
- Benchmarks: BdG at $g = 0$ (10^{-14}); ED of a 4-site ring + photon with sum rule (10^{-7}).

The energy is not the criterion

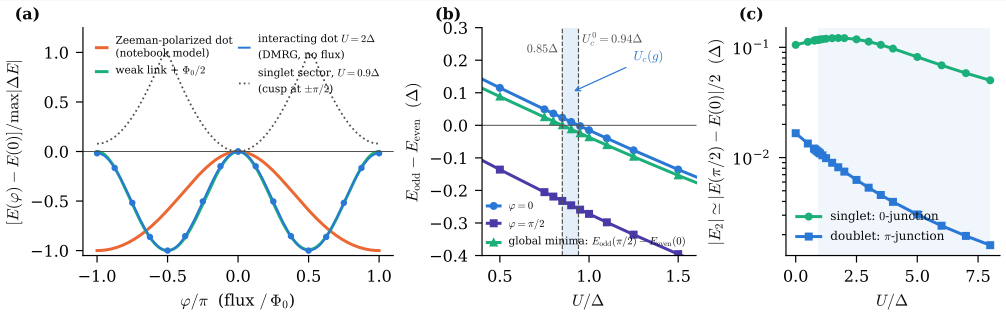
The two current states differ only by δ and the seed, $2I^\star\varphi_{c1} \sim 10^{-5}\Delta$: DMRG drifts towards the polarized state over tens of sweeps and can stop *under-polarized* within $10^{-6}\Delta$ of the ground state. (An earlier warm-start scan had this problem; the sum rule exposed it.)

Protocol for every seeded point

Two starts: (A) warm start from the vacuum side; (C) coherent state at the classical flux, annealed through seeds $10^{-2} \to 3 \cdot 10^{-3} \to 10^{-3}$. Keep the lower energy if they differ by $> 3 \times 10^{-7}\Delta$, else report a *bracket* (47/427, all inside crossovers). Accept only if the sum rule holds to $10^{-3} I_c$. Seed 10^{-4}: current unchanged above threshold, $\times 0.09$ below.

Results

The junction alone ($g = 0$): three crossings, not one



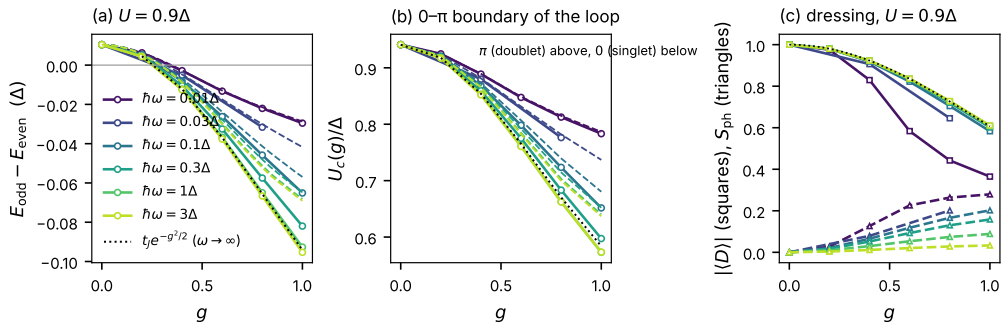
(a) The $U = 2\Delta$ doublet at *zero* flux $\equiv$ weak link + $\Phi_0/2$: $E_2 = 7.39 \times 10^{-3}\Delta$, $|E_1|/E_2 = 2\%$, $|E_4|/E_2 < 1\%$. Dotted: the singlet surface has a cusp at $\pm\pi/2$ (two even states cross).

(b) $U_c^0 = 0.94\Delta$: crossing at zero flux (the junction's own 0–π transition). 0.85Δ: crossing of the *global minima* $E_{\rm odd}(\pi/2) = E_{\rm even}(0)$. Between: the π' window [Vecino 2003; Karrasch 2008].

(c) $E_2 \propto U^{-1.3}$, approaching the $1/U$ of cotunnelling.

In the π' window a loop with a free flux switches to the doublet by letting $\Phi_0/2$ in. The mode does exactly that.

Result 1: the zero-point flux switches the loop from 0 to π



(a) $U = 0.9\Delta$, $\hbar\omega = 0.1\Delta$: $E_{\rm odd} - E_{\rm even}$ falls from $+0.010\Delta$ ($g = 0$) to -0.008Δ ($g = 0.4$) and -0.065Δ ($g = 1$). Switch at $g \simeq 0.28$, $Z \simeq 0.6\,\mathrm{k\Omega}$ – no seed, no flux, only the vacuum.

(b) $U_c(g, \omega)$ from $E_{\rm odd} = E_{\rm even}$, both sectors by full DMRG: $0.94 \to 0.83\Delta$ at $g = 0.5$; 0.65, 0.60, 0.57Δ at $g = 1$ for $\hbar\omega = 0.1$, 0.3, 1Δ (30–40%).

(c) Dressing: $|\langle D\rangle|$ follows $e^{-g^2/2}$ for $\hbar\omega \gtrsim 0.3\Delta$ (vacuum only squeezed), falls faster in the adiabatic regime (mode displaced towards the π well); entanglement $S_{\rm ph}$ largest there (0.26).

Result 1, mechanism: from the adiabatic flux average to Franck–Condon

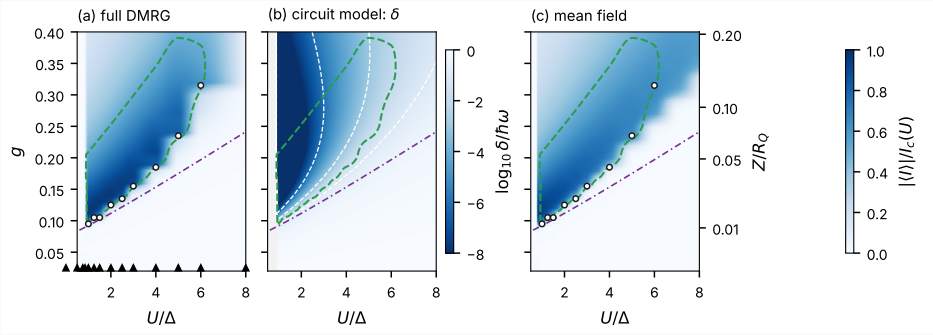
$U_c(g,\omega)/\Delta$ of the loop (full DMRG), $U_c^0 = 0.941$:

$\hbar\omega/\Delta$	$g = 0.2$	0.4	0.6	0.8	1.0
0.01	0.925	0.889	0.848	0.813	0.783
0.1	0.921	0.868	0.798	0.724	0.652
0.3	0.919	0.859	0.775	0.683	0.597
1.0	0.918	0.853	0.762	0.663	0.573
3.0	0.917	0.852	0.761	0.662	0.573
circuit model ($\hbar\omega = 0.01$)	0.925	0.889	0.849	0.815	0.787
Lang–Firsov ($\omega \to \infty$)	0.918	0.854	0.765	0.670	0.584

- $\hbar\omega \ll E_A$: exact = circuit model to $2 \times 10^{-3}\Delta$ (curvatures + relaxation towards $\Phi_0/2$; bounded by the condensation energy of the π well).
- $\hbar\omega \gtrsim \Delta$: exact = Lang–Firsov to $10^{-2}\Delta$, saturated by $\hbar\omega \simeq \Delta$ (bounded by Γ; corrections $O(t_J^2/\hbar\omega) < 0.01\Delta$).
- Not an artefact of the one-bond vertex: the neglected $O(\hbar\omega g^2/E_A)$ vanish at $\hbar\omega = 0.01\Delta$, where the shift is already half of that at 0.3Δ.
- Not mean field: HF puts the boundary at 0.82Δ for *every* g.

Prediction: a dot gate-tuned near its parity boundary, in a loop with a $Z \gtrsim 1\,\mathrm{k}\Omega$ superinductor, is driven into the π phase *by the impedance alone* – the resonator-frequency jump of [Bargerbos 2022; Pita-Vidal 2023].

Result 2 (central): the (U, g) phase diagram from full DMRG



(a) **Full DMRG**, 427 ground states: $g = 0.02$–0.40, 14 values of U (triangles), both sectors for $U \le \Delta$, seed 10^{-3}, $\chi = 300$. Colour: $|\langle I\rangle|/I_c(U)$; circles: $0.5 I_c$.

(b) Seed-free order parameter of the circuit model: tunnel splitting $\delta(U, g)$ (white: $10^{-1}, 10^{-3}, 10^{-6}\hbar\omega$); grey: singlet 0 loop. Same boundary as $\delta = 0.1\hbar\omega$.

(c) Mean field. Green dashed: $0.5 I_c$ contour of the circuit model; purple: classical $\beta = 1$. Right axis: $Z/R_Q = 4g^2/\pi$ ($g = 0.4$ is $Z = 1.3\,\mathrm{k}\Omega$).

$\hbar\omega = 7.39 \times 10^{-4}\Delta$ is chosen so that $E_J/\hbar\omega$ runs from 15 ($U \simeq \Delta$) to 2 ($U = 8\Delta$): one picture holds both the sharp transition and its quantum smearing.

Reading the phase diagram: three regions

(i) $U < U_c(g)$: the 0 loop

Singlet ground state, diamagnetic, mode *stiffened* (ω' up to 9.4ω), squeezed vacuum, current $< 10^{-2} I_c$ and linear in the seed. Boundary moves $0.94 \to 0.85\Delta$ at $g = 0.4 =$ the global-minimum crossing: the adiabatic end of Result 1.

(ii) $U > U_c$, $g > g_c(U)$: the condensate

$U = 2\Delta$: $\langle n \rangle$ $0.5 \to 7.3$ for $g = 0.11 \to 0.14$, current $0.96 I_c$, then $I \simeq \hbar\omega\varphi^\star/2g^2$ (flux saturates at $\Phi_0/2$, inductance grows). $g_c = 0.10, 0.13, 0.19, 0.24, 0.32$ at $U/\Delta = 1, 2, 4, 5, 6$: above $\beta = 1$ by $\Delta g = 0.02, 0.04, 0.12$ at $E_J/\hbar\omega = 10, 5, 3.2$; crossover widens to $\Delta g = 0.08$; $I_{\max}$ falls $0.97 \to 0.61\, I_c$.

(iii) $U = 8\Delta$ ($E_J/\hbar\omega = 2.1$): no order

β reaches 2.7 but the ground state is the cat with $\delta \simeq \hbar\omega$; current $\leq 0.04 I_c$, $\propto$ seed.

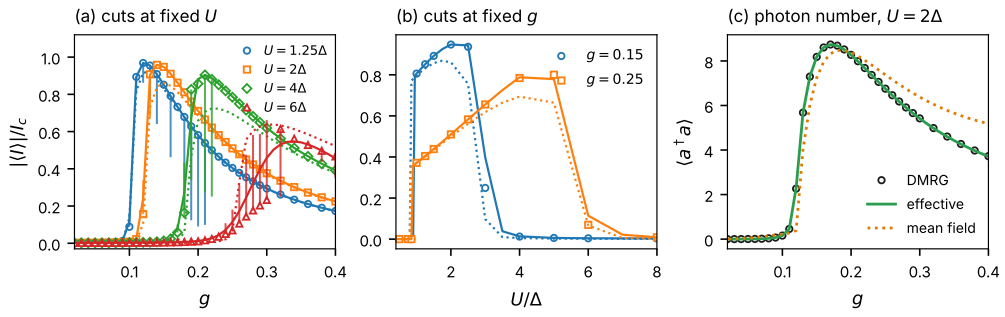
Full DMRG vs circuit model, above threshold

0.2% of I_c on average (2.4% worst), 0.1 in photon number, $10^{-5}\Delta$ in energy: at $\hbar\omega/E_A \sim 10^{-3}$ the fluxonium with the microscopic $E_s(\varphi)$ *is* the exact circuit model.

Mean field

g_c right to one grid step for $U \leq 5\Delta$, current 10–20% low, $\langle n \rangle$ high at large g (5.2 vs 3.7); orders $U = 7$–8Δ with $0.6 I_c$ (absent); orders $U = 0.85\Delta$ with $0.95 I_c$ where the exact state is the singlet 0 loop.

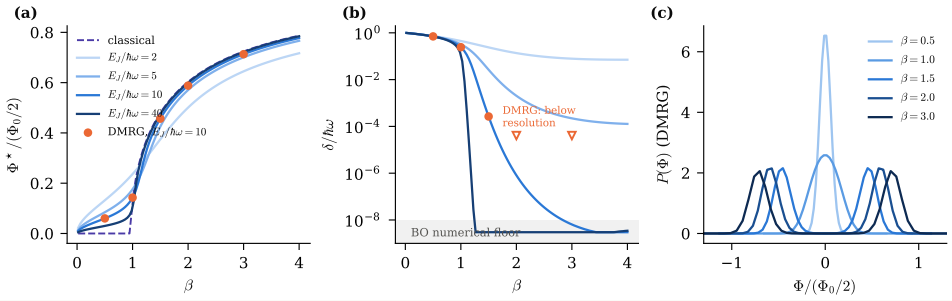
Cuts through the diagram



(a),(b) Symbols: full DMRG (bars: brackets between the two starts, all inside crossovers); solid: circuit model; dotted: mean field. The sharp step at $U = 1.25, 2\Delta$ becomes a broad crossover at $U = 6\Delta$.

(c) Photon number at $U = 2\Delta$: the condensate of $\simeq$ 7–9 photons is a *static flux* ($\langle n \rangle \simeq \langle X \rangle^2/4$), not radiation. Mean field overshoots at large g.

Result 3: the cat state and macroscopic tunnelling of a current

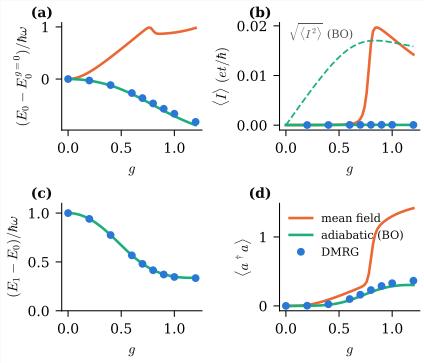


$U = 2\Delta$, $E_J/\hbar\omega = 10$, no flux, no seed: H real $\Rightarrow \langle I \rangle = 0$ in every eigenstate. (a) The flux projected on the lowest doublet has eigenvalues $\pm\Phi^\star$ ($0.59\,\Phi_0/2$ at $\beta = 2$; DMRG = circuit model to 0.2%).

(b) $\delta/\hbar\omega = 0.70,\ 0.24,\ 2.65 \times 10^{-4}$ at $\beta = 0.5, 1, 1.5$; the last one resolved directly by DMRG ($\chi = 300$ vs 400: 2.66 vs 2.65×10^{-4}): tunnelling of a seven-photon condensate, $\delta = 2 \times 10^{-7}\Delta$. Beyond: circuit model, 8×10^{-7}, 6.5×10^{-9} at $\beta = 2, 3$.

(c) The DMRG ground state *is* the cat: $|\langle\Phi\rangle| < 10^{-3}\Phi^\star$, bimodal $P(\Phi)$; a run started from the polarized state relaxes to it ($10^{-10}\Delta$). Manifold: $\{\uparrow,\downarrow\} \otimes \{\circlearrowleft,\circlearrowright\}$, split only by δ.

Where the product state fails: a cautionary tale ($E_J \simeq \hbar\omega$)



Short ring ($L = 11$), Zeeman-polarized dot, $\hbar\omega = 0.02$: the ansatz of [Varrica 2026; Andolina 2020] (and of our own first treatment).

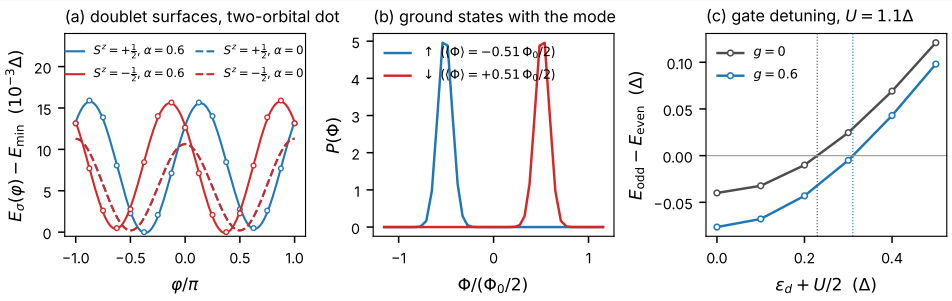
- MF: current-carrying phase for $g \gtrsim 0.75$, $\langle n \rangle \simeq 1.4$.
- DMRG: no current (linear response 7×10^{-5}), energy $1.8\hbar\omega$ *below* MF, gap stays $0.33\hbar\omega$, $\langle n \rangle \simeq 0.36$. Circuit model = DMRG.

Three reasons

(1) product state not gauge covariant; (2) $E_J \simeq \hbar\omega$: double well shallower than the zero-point energy, $P(X)$ unimodal; (3) $L \lesssim \xi$: $E_1 \gg E_2$, the instability would be of the *normal* h/e current.

Same model at $\hbar\omega = 0.002$: it does order, $g_c = 0.21$ (MF: 0.30). The control parameter is $E_J/\hbar\omega$.

With spin–orbit coupling: spin-locked currents instead of a cat



Two-orbital dot, spin-dependent phase $e^{i\alpha\sigma^z}$ on one of two interfering paths ($\alpha = 0.6$; in one channel it gauges away).
(a) Doublet surfaces $E_\sigma = E_0 + E_2\cos(2\varphi - 2\sigma\varphi_{\text{so}})$, $\varphi_{\text{so}} = 0.42$ (24°).

(b) With the mode ($\beta = 2$): the two wells of a given spin are split by $2E_2\sin 2\varphi_{\text{so}}$; $\uparrow$ sits at $-\Phi^\star$, $\downarrow$ at $+\Phi^\star$ ($\mp 0.51\,\Phi_0/2$, 6.5 photons). The manifold is one Kramers pair $\{|\uparrow\circlearrowleft\rangle, |\downarrow\circlearrowleft\rangle\}$: the inductively protected spin qubit of [del Olmo 2026].

(c) Gate: at $U = 1.1\Delta$, $\hbar\omega = 0.1\Delta$ the doublet window widens from 0.229 to 0.310Δ at $g = 0.6$ (35%; 29 μeV in Al, 0.1–0.3 mV of gate). The flux-driven switch survives: $U_c^0 = 1.36 \to 1.15\Delta$.

Where to look

Physical units: what is easy and what is hard

Flux-driven 0–π switch: easy

$\hbar\omega = 0.03$–0.3Δ, $g = 0.3$–0.6. Al ($\Delta = 180\,\mu$eV): a 1–13 GHz mode sharing its inductance with the loop, $Z = 4R_Q g^2/\pi = 0.7$–$3\,\mathrm{k}\Omega$ (granular Al, junction arrays [Manucharyan 2009; Grünhaupt 2019]), dot gate-tuned to $\sim 0.1\Delta$ of its parity boundary [Bargerbos 2022; Pita-Vidal 2023]. Readout: resonator-frequency jump; signature: driven by the impedance (superinductor length) at fixed gate.

Condensate and cat: hard

Need $E_J/\hbar\omega \gtrsim 5$ and $\hbar\omega > k_B T$. The $U = 2\Delta$, $\Gamma = \Delta/2$ junction has $E_J/h = 0.3\,\mathrm{GHz}$, $I_c = 0.65\,\mathrm{nA}$ in Al $\Rightarrow$ $\hbar\omega = E_J/10 = 32\,\mathrm{MHz}$: below the electron temperature.

U, Γ, material	E_J/h	I_c	$E_J/10h$	$E_J/10k_B$	$L_{\beta=1}$
2Δ, $\Delta/2$, Al	0.32 GHz	0.65 nA	32 MHz	1.5 mK	510 nH
2Δ, $\Delta/2$, NbTiN	2.1 GHz	4.3 nA	210 MHz	10 mK	76 nH
6Δ, 2Δ, Al	1.5 GHz	3.0 nA	150 MHz	7 mK	110 nH
6Δ, 2Δ, NbTiN	9.9 GHz	20 nA	0.99 GHz	47 mK	17 nH
6Δ, 2Δ, Nb	11.5 GHz	23 nA	1.15 GHz	55 mK	14 nH

- Strong junction ($\Gamma = 2\Delta$, $U = 6\Delta$, $E_J = 0.034\Delta$) in NbTiN/Nb: $\simeq 1\,\mathrm{GHz}$ mode, $L \simeq 17\,\mathrm{nH}$, $Z \simeq 160\,\Omega$: a π-rf-SQUID at zero flux; full DMRG: 9.6 photons, $0.94 I_c$.
- nA dot + GHz superinductor: $E_J/\hbar\omega \lesssim 3$, no current phase (the $U = 8\Delta$ line) but the cat with $\delta \sim \hbar\omega$: dip of the $0 \to 1$ frequency, jump of $\langle 0|\Phi|1\rangle$ in two-tone spectroscopy – where MF predicts $0.6 I_c$.

Summary

Take-home messages

1. Dot electrons and quantized loop flux on the same footing: full DMRG, Peierls vertex to all orders, sum rule $\langle I \rangle = \langle \Phi \rangle / L$ as acceptance test.
2. The vacuum of the mode controls the junction: its zero-point flux moves the singlet–doublet boundary by 30–40% at $g = 1$ – adiabatically through the curvatures of $E_s(\varphi)$, non-adiabatically through the Franck–Condon factor $t_J \to t_J e^{-g^2/2}$ – at impedances $Z \sim 1$–$3\,\mathrm{k}\Omega$ that exist.
3. Full-DMRG (U, g) phase diagram: the photon condensate is the screening instability $\beta > 1$ of a π loop; the exact ground state is the cat of the two current directions (δ resolved directly); zero-point motion pushes g_c above $\beta = 1$ and erases the phase for $E_J/\hbar\omega \lesssim 2.5$. In the adiabatic regime the loop *is* a fluxonium with the microscopic $E_s(\varphi)$.
4. The product-state mean field orders where the exact solution does not: at $E_J \lesssim 3\hbar\omega$ and at $U \simeq U_c$ (static moment instead of the Kramers doublet).
5. With spin–orbit the cat becomes a spin-locked persistent current (a protected spin qubit); the switch survives.

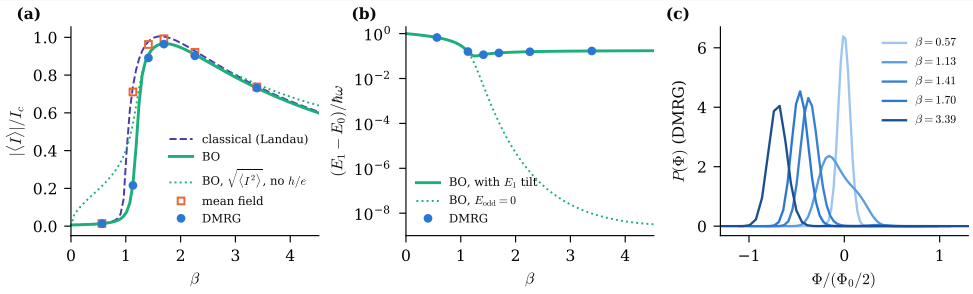
Open questions

- **Kondo regime** ($T_K \gg \Delta$): the junction returns to 0 – how does the switch look there?
- **Capacitive terms** of the displaced charge, $\hbar\omega g^2/E_A$: of order one at $\hbar\omega \gtrsim 0.3\Delta$, $g \sim 1$; a two-mode or transmission-line model would fix them.
- **Lead kinetic inductance**: with a rigid Δ all phase drops on the junction; a self-consistent BdG ring puts most of it in the leads (backup). How does it renormalize β in a real loop?
- **Quasiparticle poisoning**: a parity flip is a $\pi \leftrightarrow 0$ switch of the whole loop.
- **Dissipation and readout**: dispersive shift, $\langle 0|\Phi|1\rangle$, finite T; multi-dot junctions.

Thank you!

Backup

Backup: weak link biased by $\Phi_0/2$ ($L = 14$, $E_J/\hbar\omega = 10$)

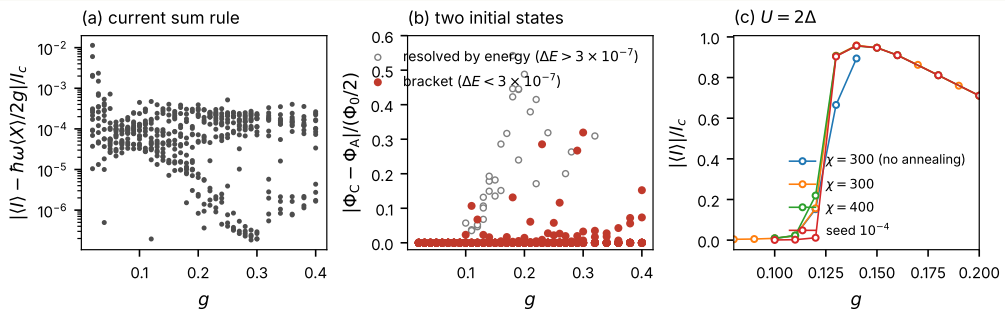


(a) Current rises at $\beta \simeq 1.1$, saturates at $0.96 I_c$. MF: 0.71 vs $0.22 I_c$ at $\beta = 1.13$, 0.2–0.4$\hbar\omega$ too high in energy. DMRG vs circuit model: $2 \times 10^{-3}\hbar\omega$, 2% of I_c.

(b) Splitting saturates: it is the tilt $2|E_1| \sin\varphi^\star \approx 0.2\hbar\omega$ of the residual h/e harmonic, not tunnelling (without it $\delta < 10^{-6}\hbar\omega$ at $\beta = 2$).

(c) $P(\Phi)$: a single displaced peak – the direction is *selected* by E_1. An intrinsic π junction removes the need for the $\Phi_0/2$ bias, not the sensitivity to a flux offset $\delta\Phi > \delta/2I^\star$.

Backup: convergence of the full DMRG on the phase diagram



(a) Sum-rule violation $|\langle I\rangle - \hbar\omega\langle X\rangle/2g|/I_c$ over the grid: median 10^{-4}, all $< 10^{-3}$ for $g \geq 0.04$.

(b) Flux difference between the two starts (vacuum side A, polarized side C): open = resolved by energy, filled = brackets (47/427).

(c) $U = 2\Delta$ near threshold: $\chi = 300$ without/with annealing, $\chi = 400$ (energies to $4\times10^{-8}\Delta$, current to 0.3%), seed 10^{-4}.

Backup: gauge covariance of the static phase; phase drop in self-consistent BdG

Non-interacting ring with a dot ($L = 14$, $t_J = 0.5$); BdG ground-state energy:

φ	one bond	uniform **A**, covariant Δ_j	uniform **A**, real Δ
0	-21.82094	-21.82094	-21.82094
0.5	-21.79739	-21.79739	-21.81251
1.0	-21.72957	-21.72957	-21.78725
$\pi/2$	-21.61009	-21.61009	-21.73791

First two agree to 10^{-10} (`tests/test_gauge_covariance.py`); a uniform **A** with a real Δ is a different model.

Weak link ($t_J = 0.5$, $L = 14$) with $\Delta_j = -U_a \langle c_{j\downarrow} c_{j\uparrow} \rangle$ solved self-consistently (complex), $|\Delta| = 1$:

pair flux 2φ	junction drop	fraction
0.4	0.110	0.275
0.8	0.221	0.277
1.2	0.335	0.279

Current conserved bond by bond; only $\approx 28\%$ of the gauge-invariant phase drops on the junction in this short ring – the leads act as a series kinetic inductance (`code/selfconsistent_phase.py`).

Backup: references

- Bulaevskii, Kuzii, Sobyanin, JETP Lett. **25**, 290 (1977); Tsuei, Kirtley, RMP **72**, 969 (2000); Ryazanov *et al.*, PRL **86**, 2427 (2001); Bauer *et al.*, PRL **92**, 217001 (2004)
- Ioffe *et al.*, Nature **398**, 679 (1999); Yamashita *et al.*, PRL **95**, 097001 (2005); Kawabata *et al.*, PRB **74**, 180502(R) (2006); Feofanov *et al.*, Nat. Phys. **6**, 593 (2010)
- Nataf, Ciuti, Nat. Commun. **1**, 72 (2010); Bamba, Inomata, Nakamura, PRL **117**, 173601 (2016); Bamba, Imoto, PRA **96**, 053857 (2017); Andolina *et al.*, PRB **100**, 121109(R) (2019); PRB **102**, 125137 (2020); Mercurio *et al.*, Phys. Rev. Res. **6** (2024)
- Glazman, Matveev, JETP Lett. **49**, 659 (1989); Spivak, Kivelson, PRB **43**, 3740 (1991); Vecino *et al.*, PRB **68**, 035105 (2003); Karrasch *et al.*, PRB **77**, 024517 (2008); Meng, Florens, Simon, PRB **79**, 224521 (2009)
- van Dam *et al.*, Nature **442**, 667 (2006); Cleuziou *et al.*, Nat. Nano. **1**, 53 (2006); Delagrange *et al.*, PRB **91**, 241401(R) (2015); Bargerbos *et al.*, PRX Quantum **3**, 030311 (2022); Hays *et al.*, Science **373**, 430 (2021); Pita-Vidal *et al.*, Nat. Phys. **19**, 1110 (2023)
- Padurariu, Nazarov, PRB **81**, 144519 (2010); del Olmo *et al.*, arXiv:2608.13530 (2026); Varrica *et al.*, arXiv:2603.07256 (2026)
- Friedman *et al.*, Nature **406**, 43 (2000); Manucharyan *et al.*, Science **326**, 113 (2009); Grünhaupt *et al.*, Nat. Mater. **18**, 816 (2019); Hwang, Puebla, Plenio, PRL **115**, 180404 (2015)
- Li *et al.*, PRB **101**, 205140 (2020); Dmytruk, Schirò, PRB **103**, 075131 (2021); Dmytruk, Trif, Simon, PRB **94**, 115423 (2016); Lang, Firsov, Sov. Phys. JETP **16**, 1301 (1963)
- Passetti *et al.*, PRL **131**, 023601 (2023); Bacciconi *et al.*, SciPost Phys. **15**, 113 (2023); Loss, Goldbart, PRB **43**, 13762 (1991); Hauschild, Pollmann, SciPost Phys. Lect. Notes 5 (2018)